# Causal Mixture-of-Experts (MoE): Semantic Step-Wise Expert Routing and Probability of Necessity and Sufficiency (PNS) for Efficient LLM Reasoning


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## Abstract & Problem Statement


Autoregressive Large Language Models (LLMs) generate tokens sequentially, applying nearly uniform computation to each token regardless of semantic role. This architectural limitation creates a severe mismatch termed **Compute-Reasoning Displacement**, where models expend comparable FLOPs on low-information connective phrases as they do on high-value logical inferences. Consequently, standard Chain-of-Thought (CoT) prompting triggers **Latency Explosion**; for example, the Qwen2.5-7B model averages 208.9 tokens per response on GSM8K, heavily dominated by repetitive scaffolding.

While sparse Mixture-of-Experts (MoE) architectures (e.g., Mixtral) optimize throughput via token-level load balancing, they are completely agnostic to the semantic structure of multi-step reasoning. To resolve this, we propose **Causal-MoE**, a framework replacing token-level routing with **semantic step-wise expert routing**. By decomposing outputs into atomic reasoning steps and dynamically assigning each to a specialized domain expert, we structurally align computational pathways with task decomposition, simultaneously eliminating reasoning opacity and massively reducing redundant token generation.


## Methodology Used: Causal-MoE Architecture

We build Causal-MoE on a 4-bit quantized Qwen2.5-7B-Instruct backbone, surgically replacing the SwiGLU-based Feed-Forward Network (FFN) sublayers at selective depths $\mathcal{L} = \{6, 12, 18, 24\}$ with custom `CausalMoEMLP` modules. This targeted injection preserves pretrained representations while balancing routing effectiveness.

The framework implements four highly specialized domain experts using Low-Rank Adaptation (LoRA) ($r = 16, \alpha = 32$): `[MATH]`, `[LOGIC]`, `[COMMONSENSE]`, and `[VERIFY]`. To retain fundamental pretrained knowledge, these experts are initialized via **Interleaved Weight Slicing**. The original FFN weights are partitioned into disjoint subsets $S_i \subset \{0, \dots, d_{FFN}-1\}$, ensuring experts inherit functionally meaningful representations. This dramatically improves convergence stability without requiring extensive, compute-heavy retraining.

### Gating Mechanism & Dynamic Routing

Each replaced FFN layer features a localized gating network evaluating the semantic tag of the active reasoning step. For a given hidden representation $h$, the routing space is mathematically defined as:

$$G(h) = hW_{gate} \in \mathbb{R}^{N_e}$$

Routing probabilities are normalized via softmax, and the top-$k = 2$ experts are deterministically selected to ensure maximum sparsity:

$$p = \text{Softmax}(G(h)), \quad (j_1, j_2) = \text{top-}k(p)$$

The final output is computed as a mathematically weighted combination of the selected expert activations:

$$\mathcal{M}\mathcal{E}(h) = \sum_{j \in \{j_1, j_2\}} \frac{p_j}{p_{j_1} + p_{j_2}} \cdot \text{Expert}_j(h)$$

## Methodology Used: PNS Optimization & Distillation

We integrate the **Probability of Necessity and Sufficiency (PNS)** framework, grounded in counterfactual causation, to explicitly quantify the causal contribution of individual reasoning steps.

To surgically eliminate verbose steps, we deploy an active **Sabotage-Based Rollout Strategy**. We inject explicit logical contradictions (e.g., "ERROR: The previous reasoning step is incorrect") into the trace and observe whether the model can recover the ground truth answer $y^*$. This enforces logical inconsistency during evaluation, providing a highly reliable necessity estimate. For a reasoning step $s_i$, the necessity score over $K = 3$ evaluations is:

$$PNS(s_i) = 1 - \frac{1}{K} \sum_{k=1}^{K} \mathbf{1}[\hat{y}_k = y^*]$$

Where $\hat{y}_k$ denotes the output obtained under the perturbed rollout. Steps with $PNS \geq 0.5$ are strictly retained. To enforce structured routing during training, we introduce the **Atomic Reasoning Trace (ART)** format where each step is explicitly prefixed with a semantic label. Training is conducted in BF16, masking cross-entropy loss over the prompt to isolate optimization purely on reasoning outputs.

## Methodology Used: The Collapsed V2 Paradigm

The **Collapsed V2 architecture** embeds the routing mechanism directly within the 7B model. This enables end-to-end generation in a single seamless forward pass. By dynamically routing computation based on semantic tags and activating appropriate expert weights natively, V2 eliminates multi-call overhead entirely. The computational cost of inference is bounded strictly by:

$$FLOP(\text{model}, T) \approx 2 \times P_{active} \times T$$

By drastically reducing both $P_{active}$ (via sparse selection) and total generated tokens $T$ (via causal distillation), V2 achieves unprecedented system-level efficiency.

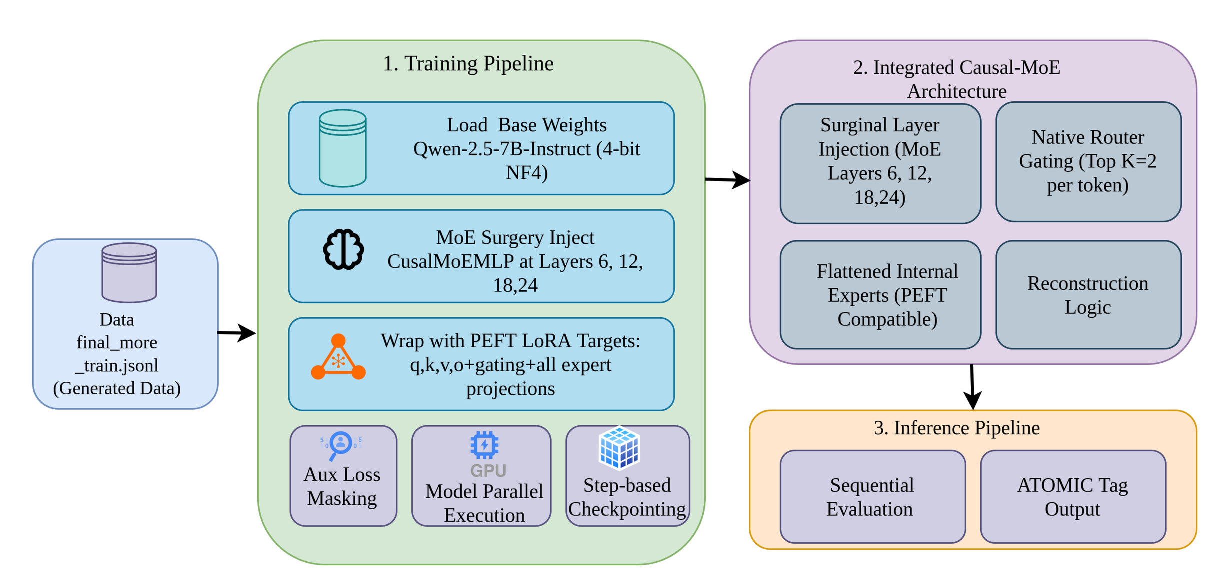


**Fig 1:** Integrated Causal-MoE V2 Architecture & Training Pipeline

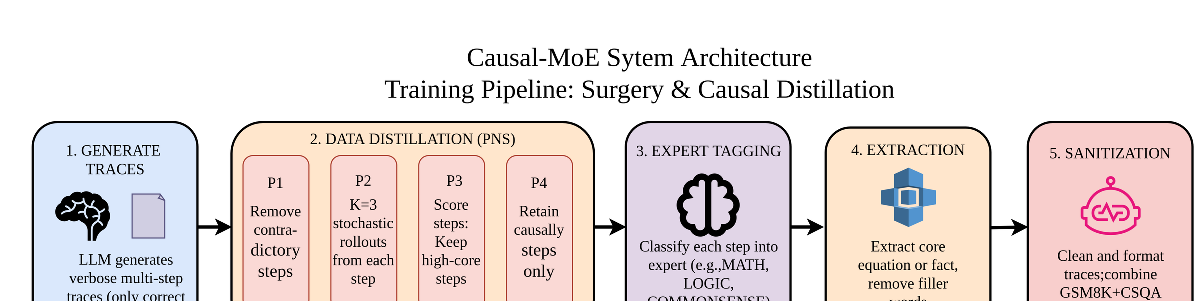


**Fig 2:** Data Distillation (PNS) & Expert Tagging Pipeline

## Obtained Results: Efficiency & Latency Breakthroughs

The central triumph of the V2 formulation is its unprecedented reduction in decoding overhead and inference latency. Evaluating on the GSM8K benchmark, the integrated V2 architecture compressed average generated tokens from 208.9 down to 75.8 (a **-63.7% reduction**). On CommonsenseQA (CSQA), token generation was compressed by an extraordinary **78.5%** (from 155.4 down to 33.4 tokens).

| Metric (GSM8K) | Base Model | Causal-MoE V2 | Reduction |
|---|---|---|---|
| Avg. Decoding Steps | 208.9 | 75.8 | -63.7% |
| Generated Tokens | 208.9 | 75.8 | **-63.7%** |
| Est. Total FLOPs | 4.28T | 2.81T | **-34.3%** |

Table 1. Computational efficiency metrics demonstrating massive overhead reduction.

Consequently, estimated total FLOPs on GSM8K plummeted from 4.28 Trillion (Base) to just 2.81 Trillion for V2. This represents a **34.3% absolute reduction** in FLOP usage per inference pass.

To quantify this hardware-to-performance synergy, we define **Reasoning Density** as $\rho_R = \frac{\text{Accuracy}}{TFLOPs}$. The V2 architecture achieved a $\rho_R$ score of 27.33 on GSM8K compared to 20.76 for the base model, yielding a highly significant **31.7% systematic improvement**. Furthermore, explicit step-wise routing proved highly beneficial for commonsense tasks, with the Causal-MoE setup achieving **83.05% accuracy** on CSQA (a +0.99 percentage point absolute improvement over base).

## Obtained Results: Comparison with State-of-the-Art

Recent methodologies (e.g., Yu et al. via DeepSeek-R1) apply PNS to prune redundant steps post-hoc on top of massive, monolithic architectures (∼671B active parameters), inherently failing to reduce the underlying computational cost per token.

Causal-MoE embeds sparse native routing directly into the architecture (7B active parameters). By executing causal logic natively, V2 achieves a comparable token reduction on GSM8K while securing a **34.3% per-pass FLOP reduction**—a vital efficiency dimension entirely unaddressed by monolithic pruning.

| Metric | DeepSeek-R1 (Yu et al.) | Causal-MoE V2 (Ours) |
|---|---|---|
| Inference Paradigm | Monolithic (Post-Hoc) | Sparse Native Routing |
| Model Scale | Dense (∼671B active) | Sparse (7B active) |
| CSQA Token Reduction | 62.8% | **78.5%** |
| GSM8K Token Reduction | 78.4% | 63.7% |
| Per-Pass FLOP Reduction | 0% | **34.3%** |
| Causal Interpretability | Implicit | **Explicit Expert Attribution** |

Table 2. Comparison of efficiency gains versus SOTA monolithic causal pruning.

## References


[1] Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, et al. Mixtral of experts. *arXiv preprint arXiv:2401.04088*, 2024.

[2] Xiao Yu, Zhaopeng Gu, and Bin Gu. Causal sufficiency and necessity improves chain-of-thought reasoning. *arXiv preprint arXiv:2506.09853*, 2025.